

Lagrange Interpolation

Interval unequal and missing value find out

∴ find the value of $f(10)$ ∴ when $y(5)=12$, $y(6)=13$, $y(9)=14$,
 $y(13)=16$.

	x	y	
x_0	5	12	y_0
x_1	6	13	y_1
x_2	9	14	y_2
x_3	13	16	y_3

$$y(10) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$= \frac{(10-6)(10-9)(10-11)}{(5-6)(5-9)(5-11)} (12)$$

$$+ \frac{(4)(1)(-1)(12)}{(-1)(-4)(-6)} +$$

$$= 14.66.$$

Numerical Integration

(website U.S.F.)

$$\textcircled{Q} \int_0^1 1 + e^{-x} \sin 4x \quad [0, 1]$$

Apply Trapezoidal Rule

$$\text{formula } \frac{h}{2} (f_0 + f_1)$$

lowest upper limit.

$$h = b - a$$

$$h = 1 - 0 = 1$$

$$f_0 = 1 + e^{-x} \sin 4x$$

$$f_0 = 1 + e^{-0} \sin 4(0)$$

$$f_0 = 1 + 0$$

$$f_0 = 1$$

$$f_1 = 1 + e^{-1} \sin 4(1) \rightarrow \text{Radian calculate.}$$

(6.28)

$$f_1 = 0.7215$$

Putting value in formulae.

$$= \frac{1}{2} (1 + 0.7215)$$

$$= 0.86075$$

Simpson Rule:

$$\text{formula: } h/3 (f_0 + f_1 + f_2)$$

$1/3$ Simpson Rule:

$$h = (b-a)/2$$

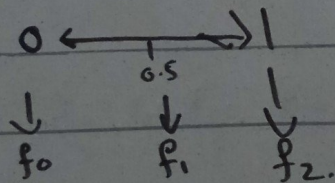
$$\text{Q) } \int_0^1 1 + e^{-x} \sin^4 x$$

cal h.

$$h = (b-a)/2$$

$$h = (1-0)/2$$

$$h = 0.5$$



$$f_0 = 1 + e^0 \sin^4(0)$$

$$f_0 = 1 + 0 = 1$$

$$f_1 = 1 + e^{-0.5} \sin^4(0.5)$$

$$f_1 = 1.5515$$

$$f_2 = 1 + e^{-1} \sin^4(1)$$

$$f_2 = 0.72159$$

$$= \frac{0.5}{3} (1 + 4(1.5515) + 0.7215)$$

$$= 1.32 \text{ ans.}$$

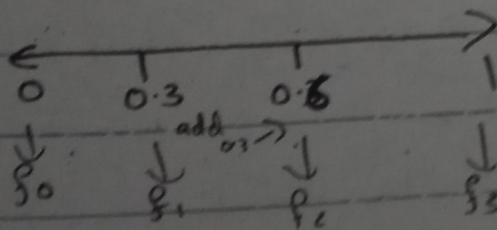
formula

$$\text{Simpson } \frac{3}{8} = \frac{3h}{8} (f_0 + 3f_1 + 3f_2 + f_3)$$

$$Q) \int_0^1 1 + e^{-x} \sin 4(x)$$

USS

$$h = \frac{b-a}{3} = \frac{1-0}{3} = 0.33$$



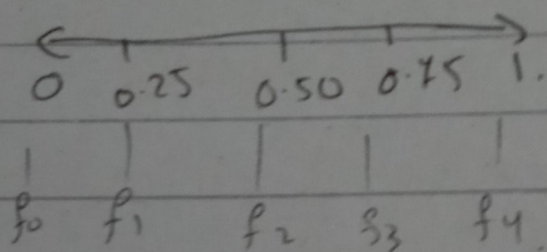
$$f_0 = 1 + e^{-}$$

Booles Rule

$$\frac{2h}{45} (7f_0 + 32f_1 + 12f_2 + 32f_3 + 7f_4)$$

$$h = \frac{b-a}{4} = \frac{1-0}{4} = 0.25$$

$$Q) \int_0^1 1 + e^{-x} \sin 4(x)$$



Differential Equation

$$\text{Euler's method (formula): } y_{n+1} = y_n + h f(x_n, y_n)$$

$$h = \frac{\text{Final value} - \text{Initial value}}{\text{step size}}$$

$$Q) f(x, y) = \frac{dy}{dx} = \frac{y-x}{y+x} \text{ with } y(0) = 1.$$

find y approximately for $x=0.1$
for Euler method.

given

$$y(0) = 1 \quad \text{it mean } (x_0, y_0) = (0, 1)$$

$$h = \frac{0.1 - 0}{5} \Rightarrow 0.02$$

step 1

$$y_{n+1} = y_n + h f(x_n, y_n)$$

for $n=0$.

$$y_{0+1} = y_0 + h f(x_0, y_0)$$

$$= 1 + 0.02 \left(\frac{0 - 1 - 0}{1 + 0} \right)$$

$$= 1.02$$

$$\text{then } x_1 = x_0 + h \Rightarrow 0 + 0.02 \Rightarrow 0.02$$

step 2

$$y_{n+1} = y_n + h f(x_n, y_n)$$

for $n=1$.

$$y_{1+1} = y_1 + h f(x_1, y_1)$$

$$y_2 = 1.02 + 0.02 \left(\frac{1.02 - 0.02}{1.02 + 0.02} \right)$$

$$\boxed{y_2 = 1.039}$$

$$x_2 = x_1 + h.$$

$$x_2 = 0.02 + 0.02 = \underline{(0.04)}$$

$$\text{Now } x_3, y_3 = ??$$

$$x_4, y_4 = ??$$

$$x_5 + y_5 = ?? \quad \text{as } 5.$$

Runge Kutta Method

$$\text{error} = e.v - \text{approx}$$

$$\text{relative error} = \frac{e.v - \text{approx}}{e.v}$$

Weddle Rule:

$$\text{formula} : \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6]$$

	x_0	x_1	x_2	x_3	x_4	x_5	
x	7.47	7.48	7.49	7.50	7.51	7.52	7.53
$f(x)$	1.93	1.95	1.98	2.01	2.03	2.06	2.09
	y_0						

$$h = \text{width of } x = 7.49 - 7.48 = 0.01.$$

$$= \frac{3 \times 0.01}{10} [1.93 + 5(1.95) + 1.98 + 6(2.01) + 2.03 + 5(2.06) + 2.09]$$

$$\underline{0.1204}$$

Gauss Quadrature Rule

$$\int_{0.1}^{1.3} 5x \cdot e^{-2x} dx$$

- Use Gauss quadrature rule to find approximate value.
- Find ~~total~~ Error when $n = 0.80386$
- Find relative Error

formula

$$\int_a^b f(x) dx = C_1 f(x_1) + C_2 f(x_2)$$

$$C_1 = \frac{b-a}{2}$$

x_1

Crank Nicolson. pdf

Final Exam

1) Lagrange interpolation (six notes). (5 marks)

2) numerical integration $\rightarrow$ 10 (marks)

$\rightarrow$ trapezium rule. - most imp. USF Website.

$\rightarrow$ $\frac{1}{3}$ Simpson Rule.

$\rightarrow$ $\frac{3}{8}$ Simpson

$\rightarrow$ bootes rule

error

relative error.

percentage error

3) Euler Method

4) Runge Kutta Method.

5) Gauss rule

6) Weddle rules

most imp.

} numerical method / six notes
USF Hollistic.

10 marks.

} six notes (5 marks).

7) Simplex Method.

8) Crank

9) Eigen values and Eigen vectors.

Why numerical integration is ^{used} taught, Applications